EXP 8

Generate Random Numbers from a Normal Distribution and Count Occurrences

Aim:

To generate random numbers from a normal distribution and count their occurrences within specified ranges.

Procedure:

1. Generate 100 random numbers using the `rnorm()` function.
2. Specify the mean and standard deviation.
3. Divide the numbers into ranges.
4. Count the occurrences in each range using `table()`.

Code:

```
set.seed(10)

numbers <- rnorm(100, mean = 0, sd
= 1)

ranges <- cut(
  numbers,
  breaks = c(-Inf, -1, 0, 1, Inf)
)

print(table(ranges))
```

Output:

```
ranges
(-Inf,-1]    (-1,0]     (0,1]  (1, Inf]
       20        36        33        11
```